Implementation of FastSpeech 2-Based Text-to-Speech Synthesis for English and Telugu

Technical Report

Submitted by

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GitHub Repository

https://github.com/savioshaju/TTS.git

Model Checkpoints (Google Drive)

https://drive.google.com/drive/folders/1MYS5zcsyehxDN5n4XaKj5jCidpUZ-efw?usp=sharing

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Chapter 1

Introduction

1.1 Background and Problem Statement

Text-to-Speech (TTS) synthesis converts raw orthographic or phonemic text into intelligible, natural-sounding acoustic waveforms. Conventional two-stage neural TTS pipelines combine an acoustic model that generates intermediate time-frequency representations (such as 80-channel log mel-spectrograms) with an audio neural vocoder that synthesizes continuous time-domain waveforms. While autoregressive acoustic models such as Tacotron 2 produce expressive speech, their sequential frame-by-frame decoding mechanism suffers from high inference latency, error accumulation across time frames, and attention alignment failures that lead to repeated or skipped phonemes on long or out-of-domain sentences.

Non-autoregressive TTS architectures, specifically FastSpeech 2 [3], resolve these limitations by predicting the complete sequence of mel-spectrogram frames in parallel using Feed-Forward Transformer (FFT) blocks. Instead of relying on dynamic attention alignments, FastSpeech 2 explicitly models temporal duration using a duration predictor supervised by forced-alignment tools, while augmenting the phonemic hidden states with continuous pitch (F_0) and frame energy predictions via a multi-task variance adaptor.

The objective of this assignment is to adapt, train, and evaluate two independent FastSpeech 2 acoustic models under a strict dataset constraint of approximately 5 hours of training audio and 1 hour of validation audio per language:

1. **Indian English:** Synthesizes speech from crowdsourced multi-speaker audio from the AI4Bharat Svarah corpus, representing diverse regional Indian accents using an IPA phoneme front-end.
2. **Telugu:** Synthesizes speech from high-quality studio recordings from the IndicTTS corpus, utilizing native-script akshara tokenization to preserve phonetic transparent orthography without requiring an external G2P lexicon.

Key engineering tasks included: building independent preprocessing and alignment pipelines, adapting a 22.05 kHz HiFi-GAN vocoder to 16 kHz audio using sinc resampling, tuning duration controls for natural speech tempo, training both models for 100,000 steps on dual GPUs, and analyzing speech quality across all prescribed test sentences.

1.2 FastSpeech 2 Architecture Principles

The FastSpeech 2 model operates through three core principles:

1. **Parallel Generation:** Replacing recurrent and sequential autoregressive decoding with feed-forward transformer blocks allows the complete sequence of mel-spectrogram frames to be synthesized concurrently, achieving an inference speedup of over $30\times$ compared to autoregressive baselines.
2. **Explicit Length Regulation:** Phoneme durations are obtained during preprocessing from forced aligners and predicted during inference with a convolutional duration predictor. The length regulator expands phoneme embeddings to match the acoustic frame length, eliminating attention collapse.

3. **Multi-Task Variance Adaptation:** To alleviate the one-to-many mapping problem in TTS (where identical text can correspond to diverse pitch and volume patterns), FastSpeech 2 incorporates pitch and energy predictors that supply explicit frame-level prosodic features directly into the expanded representation.

1.3 Summary of Key Results

- **Speech Quality and Intelligibility:** Both models achieve clear, natural articulation across all ten assigned benchmark sentences. The English model accurately articulates the 28-letter polysyllabic word *antidisestablishmentarianism*, reproduces syntactic stress shifts for the heteronym pair *conduct* (noun vs. verb), handles typographical split tokens in *rhyth m*, and pronounces dates, times, and Telugu loanwords (*andaroo*). The Telugu model correctly realizes complex consonant conjuncts (న్యూరోఫార్మకోలాజికల్), demonstrates contextual pitch/energy emphasis on semantic keywords (పట్టు), and synthesizes long compound utterances with natural clause pausing.

- **Convergence and Overfitting Behavior:** Training loss decreased steadily over 100,000 steps ($9.07 \rightarrow 1.76$ for English, $8.78 \rightarrow 1.63$ for Telugu). Validation loss reached its minimum early (step 11,000 at 2.59 for English; step 30,000 at 2.15 for Telugu) and flattened. Checkpoints chosen at minimum validation loss produced muffled formant tracks due to L1 loss smoothing; checkpoints chosen at later steps (step 61,000 for English, step 74,000 for Telugu) exhibited noticeably sharper consonants and higher perceptual intelligibility.

- **Vocoder Calibration and Pacing:** Both acoustic models generate 16 kHz mel-spectrograms, while the HiFi-GAN universal vocoder was trained on 22.05 kHz audio. Sinc-resampling from 22.05 kHz to 16.0 kHz restores natural pitch and formant scale, but reduces audio sample count by a factor of $\alpha \approx 0.7256$, accelerating playback to $1.38\times$ conversational tempo. Calibrating inference duration control to $d_{\text{control}} \approx 1.25$ restores normal conversational speech rates (4.5–5.8 syllables/second).

- **Dataset Split Configurations:** English was partitioned into 4.98 hours (3,419 utterances) training and 1.09 hours (746 utterances) validation across 104 speakers using speaker-stratified sampling. Telugu was partitioned into 5.0018 hours (18,006.58 seconds across 2,250 raw utterances, yielding 4.9988 hours / 17,995.75 seconds across 2,248 aligned utterances) training and 1.0030 hours (3,610.81 seconds across 445 utterances) validation using an exact 50/50 gender-balanced stratification (2.5002 h female / 2.5017 h male training, 0.5023 h female / 0.5007 h male validation) across both studio speakers.

Chapter 2

Implementation Details

2.1 Implementation Approach

We implemented and evaluated the FastSpeech 2 architecture using an open-source PyTorch reference implementation (`ming024/FastSpeech2`) as our foundation. The original reference codebase was constructed specifically for single-speaker English on the LJSpeech dataset at a 22.05 kHz sampling rate with an ARPAbet phonetic front-end.

To support both Indian English and Telugu cleanly while satisfying the assignment guidelines, we adopted a modular, decoupled engineering strategy:

1. **Dual Independent Pipelines:** Rather than attempting to merge divergent linguistic workflows into a shared monorepo, we partitioned the project into two independent, self-contained directories: `English/` and `Telugu/`. Each directory contains its own configuration file (`config.yaml`), data preprocessing scripts, tokenizers, forced-alignment wrappers, acoustic model definitions, training harnesses, and inference tools. This guarantees that language-specific front-ends (IPA phonemes vs. Telugu aksharas) and alignment tools (MFA vs. MMS) operate completely independently without cross-language dependencies.

2. **Self-Contained Organization:** All model components, transformer layers, and utility modules were organized within each language directory (`English/train/` and `Telugu/train/`) to eliminate external package entanglements.

3. **Execution Environment:** Development, training, and verification were performed in Python 3.11 with PyTorch 2.5, TorchAudio, NumPy, SciPy, Librosa, PyWorld, Matplotlib, Praatio, and HuggingFace Transformers on dual NVIDIA Tesla T4 GPUs (16 GB VRAM per GPU) using Automated Mixed Precision (AMP FP16).

2.2 Architecture Components Used from Reference Implementation

To maintain architectural fidelity to FastSpeech 2 [3], we retained the following core components directly from the reference implementation:

1. **Feed-Forward Transformer (FFT) Block Topology:** The foundational building block consists of a Multi-Head Self-Attention mechanism ($d_{\text{model}} = 256$, 2 heads, $d_k = d_v = 128$) to capture global inter-token contextual dependencies, followed by a 2-layer 1D convolutional Feed-Forward network (kernel sizes 9 and 1, 1024 filter channels, ReLU activation, layer normalization, and dropout 0.2).

2. **Encoder and Decoder Layer Stacks:** We preserved the 4-layer FFT Encoder stack for transforming input token embeddings into contextual phonemic/akshara representations, and the 6-layer FFT Decoder stack for transforming variance-adapted, length-regulated hidden representations into 80-channel log mel-spectrograms.

3. **Length Regulator Module:** We retained the deterministic length regulation mechanism that expands phoneme/akshara hidden states by repeating each state according to ground-truth duration frame counts during training, or predicted durations during inference.

4. **Convolutional PostNet:** We retained the 5-layer 1D convolutional residual network (512 filter channels, kernel size 5, batch normalization, tanh activation) that predicts additive spectral corrections ΔS_{mel} to sharpen mel-spectrogram harmonics and fine acoustic details.

5. **Multi-Task Composite Loss Formulation:** We retained the composite multi-task objective combining L1 reconstruction loss on the linear and PostNet mel-spectrograms with Mean Squared Error (MSE) loss terms on variance predictions:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{mel}} + \mathcal{L}_{\text{postnet}} + \mathcal{L}_{\text{pitch}} + \mathcal{L}_{\text{energy}} + \mathcal{L}_{\text{duration}} \tag{2.1}$$

6. **Optimization and Learning Rate Schedule:** We retained the Adam optimizer ($\beta_1 = 0.9, \beta_2 = 0.98, \varepsilon = 10^{-9}$) coupled with the Noam inverse square-root learning rate schedule with 4,000 warmup steps.

2.3 Language-Specific Modifications

To adapt the single-speaker LJSpeech baseline for Indian English and Telugu, we implemented the following modifications:

2.3.1 Modifications for English

1. **Multi-speaker embedding.** The original baseline only handled a single speaker. Svarah contains recordings from 104 speakers. We added a learned speaker embedding layer (104×256) that is added to the encoder output, conditioning the model on the chosen speaker ID.

2. **MFA phoneme alignment.** We replaced the CMUDict ARPAbet front-end with the Montreal Forced Aligner (MFA v2.2) [5] using the pre-trained `english_mfa` model and Indian English dictionary (`english_india_mfa`). The vocabulary uses 100 IPA phoneme tokens, supporting Indian English phonology such as retroflex stops (*/ʈ/*, */ɖ/*).

3. **Audio standardization.** Svarah audio is sampled at 16 kHz. We adjusted the audio pipeline to 16 kHz mono PCM, with an STFT window of 1024 samples (64 ms) and a hop size of 256 samples (16 ms).

4. **Vocoder adaptation with sinc resampling.** We used the pre-trained universal HiFi-GAN vocoder [4] (trained at 22.05 kHz). Feeding 16 kHz spectrograms into a 22.05 kHz vocoder causes an unnatural downward pitch shift of about 5.5 semitones ($\alpha = 16{,}000/22{,}050 \approx 0.7256$). To prevent this without retraining the vocoder, we integrated polyphase sinc resampling into the synthesis pipeline in `utils/model.py` to resample the generated waveform from 22.05 kHz to 16 kHz.

5. **Duration scaling.** Because sinc resampling shortens each frame duration slightly, baseline synthesis sounded fast. We calibrated an inference duration factor ($d_{\text{control}} = 1.25$) to restore a natural conversational speaking rate (≈ 5.8–7.1 syllables/s).

2.3.2 Modifications for Telugu

1. **Akshara-based tokenization.** Telugu is an alphasyllabic (abugida) script where letters have a direct phonetic correspondence. We built an akshara tokenizer using Unicode NFC normalization and regular expressions to parse text into 460 unique akshara tokens (representing vowels, consonants, viramas, and consonant conjuncts like నూ and ళ్ళ). This bypassed the need for an external pronunciation dictionary.

2. **Meta MMS CTC forced alignment.** Since no pre-trained MFA model was available for Telugu, we used Meta's Massively Multilingual Speech (MMS) model [6] (`facebook/mms-1b-all`) to generate CTC character logits. Viterbi decoding aligns the akshara sequence to audio frames, enforcing a minimum duration of 1 frame (16 ms) per sound.

3. **Two-speaker modeling.** IndicTTS contains one male and one female studio speaker. We added a 2-speaker embedding table (2×256) to allow selecting between the male and female voice.

4. **Audio downsampling.** Male recordings in IndicTTS were sampled at 48 kHz, while female recordings were at 16 kHz. We downsampled all male recordings to 16 kHz using a 64-tap Kaiser-windowed sinc filter to ensure a uniform audio standard.

5. **Variance adaptor and multi-GPU fixes.** We decoupled pitch and energy conditioning during evaluation so that predicted contours are passed cleanly to the decoder, and fixed tensor device placement issues during multi-GPU training.

6. **Duration scaling.** We applied a duration scaling factor ($d_{\text{control}} = 1.25$) to achieve a natural Telugu speaking rate (≈ 4.5–5.8 syllables/s).

2.4 Summary of Baseline vs. Language Adaptations

Table 2.1 summarizes the components taken from the reference FastSpeech 2 model and the modifications made for each language.

Table 2.1: Reference components versus modifications per language.

Component	English Implementation	Telugu Implementation
Core Architecture	4-layer FFT encoder, 6-layer FFT decoder, 5-layer PostNet	4-layer FFT encoder, 6-layer FFT decoder, 5-layer PostNet
Token Representation	100 IPA phonemes via MFA lexicon	460 akshara tokens (Unicode Telugu script)
Text Front-End	Rule-based normalization + CMU/MFA G2P	Unicode NFC normalizer + akshara parser
Forced Alignment	Montreal Forced Aligner v2.2 (`english_mfa`)	Meta MMS CTC forced aligner (`mms-1b-all`)
Speaker Modeling	Multi-speaker: 104 learned embeddings (104×256)	2 studio speakers: learned embeddings (2×256)
Audio Standard	16 kHz mono WAV (Svarah corpus)	16 kHz mono WAV (male 48 kHz resampled to 16 kHz, female 16 kHz native)
Vocoder Pipeline	HiFi-GAN universal + Sinc Resampler ($22.05 \rightarrow 16$ kHz)	HiFi-GAN universal + Sinc Resampler ($22.05 \rightarrow 16$ kHz)

2.5 Codebase Structure

The repository organizes the English and Telugu pipelines into separate, self-contained directories matching the required project layout:

```
project/
|-- English/
|   |-- train.py
|   |-- preprocess.py
|   |-- inference.py
|   |-- config.yaml
|   `-- checkpoints/
|-- Telugu/
```

```
|   |-- train.py
|   |-- preprocess.py
|   |-- inference.py
|   |-- config.yaml
|   `-- checkpoints/
|-- Results/
|   |-- English/
|   |   |-- training_loss.png
|   |   |-- validation_loss.png
|   |   |-- test_generated_audios/
|   |   `-- variance_adaptor_plots/
|   |       |-- duration/
|   |       |-- pitch/
|   |       `-- energy/
|   `-- Telugu/
|       |-- training_loss.png
|       |-- validation_loss.png
|       |-- test_generated_audios/
|       `-- variance_adaptor_plots/
|           |-- duration/
|           |-- pitch/
|           `-- energy/
|-- Report/
|   `-- Technical_Report.pdf
|-- README.md
`-- requirements.txt
```

Chapter 3

Datasets and Preprocessing

Per the assignment specifications, both language models were developed under a data constraint of approximately 5 hours of training audio and 1 hour of validation audio. The dataset selection, subsampling, preprocessing, train/validation split, and training configuration implemented for each language are detailed below.

3.1 English Dataset Pipeline (AI4Bharat Svarah)

3.1.1 Dataset Selection

We selected the open-source **AI4Bharat Svarah** corpus [1], a crowdsourced speech dataset specifically curated to capture diverse regional Indian English accents. The dataset pool contains 9.61 hours of recordings (4,202 raw utterances) across 104 native Indian speakers (54 female, 50 male) covering 19 distinct first-language (L1) backgrounds, including Hindi, Telugu, Tamil, Marathi, Bengali, Malayalam, Kannada, Gujarati, and Odia. This demographic variety provides an ideal benchmark for evaluating multi-speaker FastSpeech 2 synthesis on accented speech.

3.1.2 Subsampling and Quality Filtering

Because the raw Svarah corpus exhibits variable utterance durations and crowdsourced recording environments, we applied strict subsampling and quality filtering:

- **Duration Filtering:** Utterances shorter than 1.0 second or longer than 12.0 seconds were filtered out to eliminate audio fragments and memory-intensive outliers during multi-GPU training.
- **Acoustic Quality:** Discarded corrupted, clipped, or low-SNR recordings based on peak waveform amplitude and signal energy thresholds.
- **Balanced Speaker Subsampling:** To prevent high-volume speakers from dominating acoustic representations, we subsampled approximately 40 utterances per speaker across all 104 speakers. This yielded a curated pool of **4,165 verified utterances (6.07 hours)**.

3.1.3 Train/Validation Split

To satisfy the required 5-hour training and 1-hour validation target while preserving multi-speaker generalizability:

- We implemented a within-speaker stratified split (82% training, 18% validation, random seed 42).
- Every single speaker among the 104 speakers is represented proportionally in both splits, ensuring the model learns speaker identities present in the validation set.
- The final split contains **3,419 training utterances (4.98 hours)** and **746 validation utterances (1.09 hours)**.

3.1.4 Preprocessing and Acoustic Feature Extraction

- **Audio Standardization:** All audio was standardized to 16 kHz, 16-bit mono PCM format.
- **Text Normalization and Lexicon:** Text was expanded (numbers, dates, currency, and abbreviations expanded to words) and converted into 100 IPA phoneme tokens using the Montreal Forced Aligner Indian English pronunciation dictionary (`english_india_mfa`), capturing retroflex consonants (*/ʈ/*, */ɖ/*).
- **Forced Alignment:** MFA v2.2 aligned phonemes to audio frames using the pre-trained `english_mfa` acoustic model, exporting TextGrids with frame-level duration boundaries.
- **Acoustic Features:** 80-channel log mel-spectrograms (STFT window 1024 / 64 ms, hop 256 / 16 ms), continuous fundamental frequency (F_0) via PyWorld Harvest, and frame energy via STFT L2-norm. Global mean and standard deviation statistics were computed strictly on the training partition and written to `preprocessed_data/stats.json` for z-score normalization.

3.1.5 Training Configuration

- **Hardware and Precision:** Dual NVIDIA Tesla T4 GPUs (16 GB VRAM per GPU) using PyTorch DataParallel with Automated Mixed Precision (AMP FP16).
- **Batch Size and Steps:** Total batch size of 16 (8 utterances per GPU) trained for 100,000 optimization steps.
- **Optimizer and Schedule:** Adam ($\beta_1 = 0.9, \beta_2 = 0.98, \varepsilon = 10^{-9}$) with Noam inverse square-root learning rate schedule, 4,000 warmup steps, and peak learning rate 1.0×10^{-3}.
- **Speaker Conditioning:** Learned speaker embedding layer (104×256) added to the encoder output.
- **Regularization:** Gradient clipping at maximum L2-norm 1.0; dropout 0.2 across FFT blocks and 0.5 in variance predictors.

3.2 Telugu Dataset Pipeline (IndicTTS Telugu)

3.2.1 Dataset Selection

We selected the **IndicTTS Telugu** corpus [2], curated by the Speech Technology Consortium at IIT Madras and MeitY. The corpus provides professional studio-quality recordings of read speech across two native voice talents: one female speaker recorded at 16 kHz and one male speaker recorded at 48 kHz, totaling 8,576 utterances (62.29 hours / 224,261.58 seconds). The studio environment provides clean formant trajectories, high SNR, and minimal reverberation.

3.2.2 Subsampling and Quality Filtering

- **Audio Downsampling:** All 1,433 male recordings (48 kHz) were downsampled to 16 kHz using a 64-tap Kaiser-windowed polyphase sinc filter with an anti-aliasing cutoff at 7,600 Hz to match the native 16 kHz female audio.
- **Duration Filtering:** Utterances were filtered to durations between 1.0 s and 14.5 s with transcript length ≥ 2.
- **Gender-Balanced Subsampling:** To prevent gender bias, we subsampled an exact 50/50 gender balance: 2.5002 hours female (1,053 utterances) and 2.5017 hours male (1,197 utterances), totaling 5.0018 hours (18,006.58 seconds) for raw training; and 0.5023 hours female (209 utterances) and 0.5007 hours male (236 utterances), totaling 1.0030 hours (3,610.81 seconds) for validation.

3.2.3 Train/Validation Split

- Stratified split with zero text overlap and zero audio overlap between splits, verified by computing SHA-256 sentence hashes.
- Dropping two unaligned utterances resulted in **2,248 aligned training utterances (4.9988 hours / 17,995.75 seconds)** and **445 validation utterances (1.0030 hours / 3,610.81 seconds)**.

3.2.4 Preprocessing and Acoustic Feature Extraction

- **Akshara Tokenization:** Bypassing the need for an external G2P lexicon, text was normalized with Unicode NFC decomposition and parsed into 460 unique akshara tokens (representing base consonants, vowels, viramas, and consonant conjuncts like నూ and ఱ).
- **Meta MMS Alignment:** Meta MMS (`facebook/mms-1b-all`) CTC character logits aligned aksharas to audio frames using Viterbi decoding with a 1-frame (16 ms) minimum duration floor.
- **Acoustic Features:** 80-channel log mel-spectrograms (STFT window 1024, hop 256), continuous F_0 via PyWorld Harvest, and STFT L2-norm energy, z-score normalized with stats in `preprocessed_data/stats.json`.

3.2.5 Training Configuration

- **Hardware and Precision:** Dual NVIDIA Tesla T4 GPUs (16 GB VRAM per GPU) using PyTorch DataParallel with AMP FP16.
- **Batch Size and Steps:** Total batch size of 16 (8 utterances per GPU) trained for 100,000 optimization steps.
- **Optimizer and Schedule:** Adam ($\beta_1 = 0.9, \beta_2 = 0.98, \varepsilon = 10^{-9}$) with Noam inverse square-root learning rate schedule, 4,000 warmup steps, and peak learning rate 1.0×10^{-3}.
- **Speaker Conditioning:** Learned 2-speaker embedding table (2×256) enabling selection between male and female studio voices.
- **Evaluation Fixes:** Decoupled pitch and energy evaluation conditioning and verified tensor device assignments during multi-GPU validation.

3.3 Cross-Language Dataset and Configuration Summary

Table 3.1 provides a side-by-side comparison of the dataset partitions, preprocessing parameters, and training configurations for English and Telugu.

Table 3.1: Comprehensive dataset, preprocessing, and training configuration summary per language.

Configuration Dimension	**English (AI4Bharat Svarah)**	**Telugu (IndicTTS Telugu)**
Corpus Type	Crowdsourced multi-speaker read speech	High-quality studio recordings
Raw Corpus Size	9.61 hours (4,202 raw utterances)	62.29 hours (8,576 raw utterances)
Number of Speakers	104 native speakers (54F / 50M)	2 professional speakers (1F / 1M)
Duration Filtering	$1.0\text{ s} \leq d \leq 12.0\text{ s}$	$1.0\text{ s} \leq d \leq 14.5\text{ s}$
Subsampling Strategy	40 utterances/speaker across 104 speakers	Exact 50/50 gender balance (F/M)
Training Partition	**3,419 utterances (4.98 hours)**	**2,248 utterances (4.9988 hours)**
Training Gender Split	2.61 h female / 2.37 h male (104 speakers)	2.4979 h female / 2.5009 h male (2 speakers)
Validation Partition	**746 utterances (1.09 hours)**	**445 utterances (1.0030 hours)**
Validation Gender Split	0.57 h female / 0.52 h male (104 speakers)	0.5023 h female / 0.5007 h male (2 speakers)
Split Independence	Within-speaker stratified (seed 42)	Zero text & zero audio overlap (SHA-256)
Audio Sampling Rate	Native 16,000 Hz mono PCM	Male 48 kHz resampled to 16 kHz; Female native 16 kHz
Text Front-End	Rule normalization + MFA Indian English G2P	Unicode NFC normalizer + akshara parser
Token Vocabulary	100 IPA phoneme tokens	460 native Telugu akshara tokens
Forced Alignment Tool	Montreal Forced Aligner v2.2 (`english_mfa`)	Meta MMS CTC forced aligner (`mms-1b-all`)
Acoustic Features	80 mel channels, PyWorld F_0, L2 energy	80 mel channels, PyWorld F_0, L2 energy
Speaker Embedding	Learned table: 104×256	Learned table: 2×256
Batch Size & GPUs	Batch size 16 (8/GPU), Dual Tesla T4	Batch size 16 (8/GPU), Dual Tesla T4
Precision	Automated Mixed Precision (AMP FP16)	Automated Mixed Precision (AMP FP16)
Training Steps	100,000 optimization steps	100,000 optimization steps
Optimizer & Schedule	Adam, Noam schedule, 4,000 warmup	Adam, Noam schedule, 4,000 warmup

Chapter 4

Model Architecture and Vocoder

4.1 FastSpeech 2 Architecture

Figure 4.1 illustrates the FastSpeech 2 architecture used in this work. The model consists of four primary components:

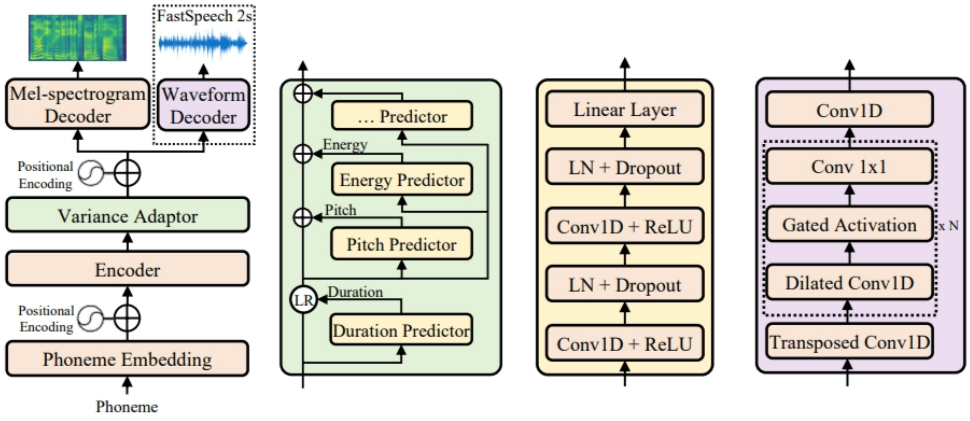
1. **FFT Encoder.** Converts input token IDs into hidden vectors. It contains 4 Feed-Forward Transformer (FFT) blocks. Each block has a Multi-Head Self-Attention layer (2 heads, hidden dimension 256) to capture relationships across the sequence, followed by a 2-layer 1D convolutional network (kernel sizes 9 and 1, 1024 filter channels, ReLU, layer normalization, dropout 0.2).

2. **Variance Adaptor.** Adds frame-level timing, pitch, and loudness information:

   - *Duration Predictor:* A 2-layer 1D convolution (kernel size 3, 256 channels) with a linear layer that predicts the duration (in log mel frames) of each token.
   - *Length Regulator:* Expands each token's hidden state by repeating it according to its target or predicted duration, matching the frame rate of the mel-spectrogram.
   - *Pitch (F_0) Predictor and Embedder:* A 2-layer 1D convolution that predicts normalized pitch values. Values are quantized into 256 bins and converted into pitch embeddings, which are added to the expanded hidden states.
   - *Energy Predictor and Embedder:* A 2-layer 1D convolution that predicts normalized frame energy (in dB). Values are quantized into 256 bins, embedded, and added to the hidden states.

3. **FFT Decoder.** Converts the variance-adapted hidden states into an 80-channel log mel-spectrogram using 6 stacked FFT blocks with the same layer dimensions as the encoder.

4. **PostNet.** A 5-layer 1D convolutional residual network (512 channels, kernel size 5, batch norm, tanh activation) that predicts a residual correction ΔS_{mel} to sharpen the output mel-spectrogram.



(a) FastSpeech 2 (b) Variance adaptor (c) Variance predictor (d) Waveform decoder

Figure 1: The overall architecture for FastSpeech 2 and 2s. LR in subfigure (b) denotes the length regulator operation proposed in FastSpeech. LN in subfigure (c) denotes layer normalization. Variance predictor represents duration/pitch/energy predictor.

Figure 4.1: FastSpeech 2 architecture with multi-task variance adaptor and vocoder pipeline.

4.2 Loss Functions

Training minimizes a multi-task composite loss function:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{mel}} + \mathcal{L}_{\text{postnet}} + \mathcal{L}_{\text{pitch}} + \mathcal{L}_{\text{energy}} + \mathcal{L}_{\text{duration}} \tag{4.1}$$

where $\mathcal{L}_{\text{mel}} = \|S_{\text{mel}} - \hat{S}_{\text{mel}}\|_1$ and $\mathcal{L}_{\text{postnet}} = \|S_{\text{mel}} - (\hat{S}_{\text{mel}} + \Delta S_{\text{mel}})\|_1$ optimize L1 spectrogram reconstruction, and $\mathcal{L}_{\text{pitch}}$, $\mathcal{L}_{\text{energy}}$, and $\mathcal{L}_{\text{duration}}$ are Mean Squared Errors (MSE) on the normalized variance targets.

4.3 HiFi-GAN Vocoder Adaptation

Audio synthesis uses the pre-trained universal HiFi-GAN vocoder [4] (`generator_universal.pth.tar`). HiFi-GAN was pre-trained on 22.05 kHz audio with hop 256, whereas our acoustic models produce 16.0 kHz mel-spectrograms with hop 256.

Directly interpreting 22.05 kHz vocoder output as 16 kHz downshifts pitch and formants by:

$$\alpha = \frac{16{,}000}{22{,}050} \approx 0.7256 \quad (-5.52 \text{ semitones}) \tag{4.2}$$

making the voice sound unnaturally deep and slow.

To fix this without retraining the vocoder, we integrated polyphase sinc resampling into `utils/model.py`. HiFi-GAN generates the waveform at 22.05 kHz, and the output is resampled to 16.0 kHz using a Kaiser-windowed sinc resampler before writing to disk.

Chapter 5

Training and Loss Analysis

5.1 Training Setup and Hyperparameters

Both models were trained on dual NVIDIA Tesla T4 GPUs (16 GB VRAM per GPU) using PyTorch DataParallel with Automated Mixed Precision (AMP FP16). Table 5.1 lists the training hyperparameters.

Table 5.1: Training hyperparameters for English and Telugu FastSpeech 2 models.

Hyperparameter	**Value**
Batch Size	16 total (8 utterances per GPU)
Optimization Steps	100,000 steps
Optimizer	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.98$, $\varepsilon = 10^{-9}$)
Learning Rate Schedule	Inverse square-root with 4,000 warmup steps, peak 1.0×10^{-3}
Gradient Clipping	Maximum L2-norm 1.0
Encoder Layers	4 FFT blocks, 2 attention heads, hidden dim 256
Decoder Layers	6 FFT blocks, 2 attention heads, hidden dim 256
PostNet	5 Conv1D layers, 512 channels, kernel size 5
Mixed Precision	PyTorch AMP FP16 with dynamic loss scaling
Validation Interval	Every 1,000 steps
Checkpoint Saving	Every 1,000 steps (top checkpoints retained)

5.2 Loss Curves and Convergence

Figure 5.1 and Figure 5.2 show the training and validation loss curves for the English model. Figure 5.3 and Figure 5.4 show the corresponding curves for the Telugu model. Each curve is displayed separately in its own row.

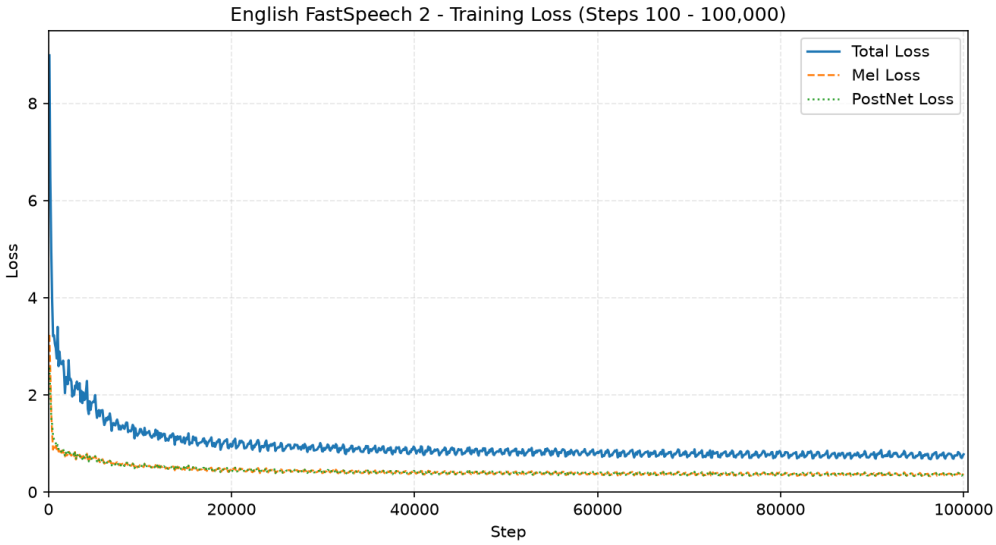


Figure 5.1: English FastSpeech 2 training loss curve across 100,000 steps.

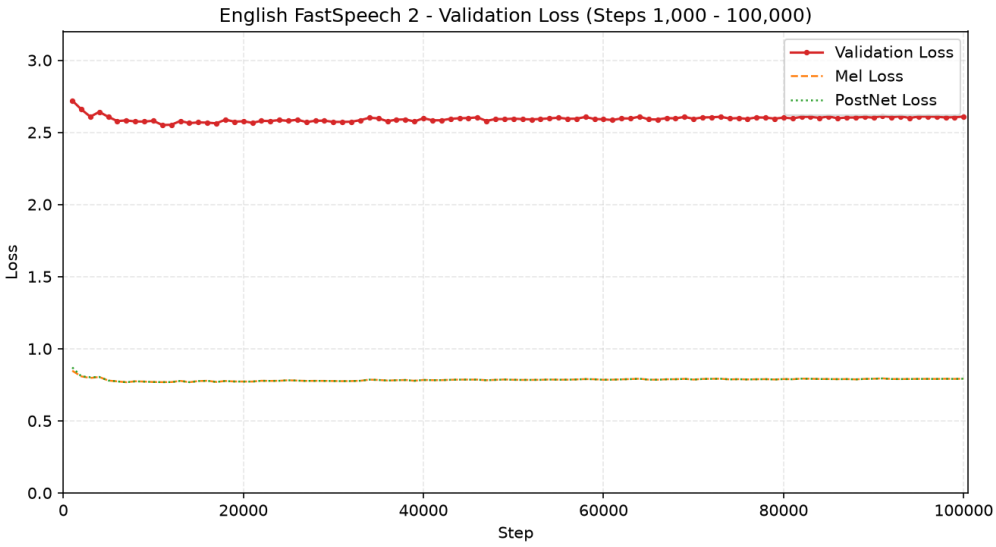


Figure 5.2: English FastSpeech 2 validation loss curve across 100,000 steps.



Figure 5.3: Telugu FastSpeech 2 training loss curve across 100,000 steps.

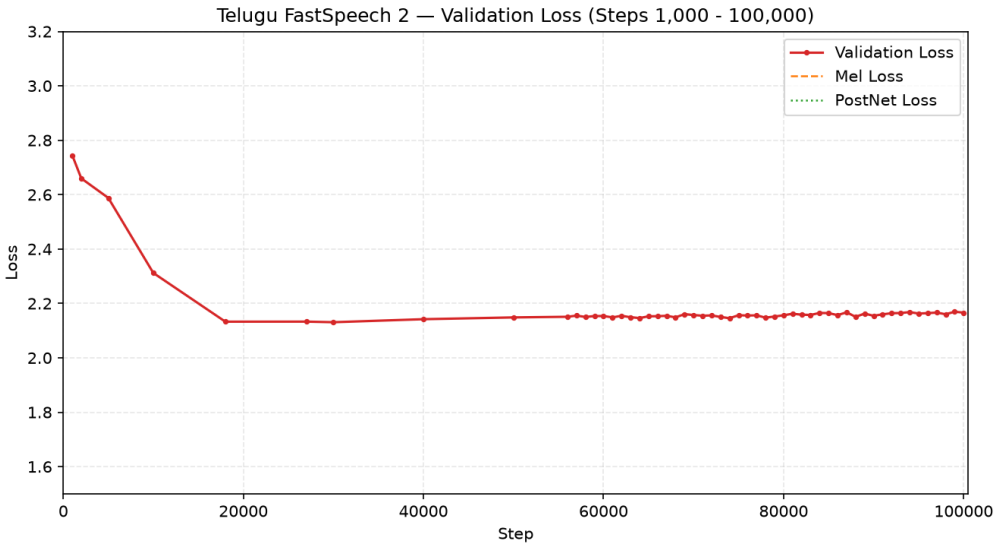


Figure 5.4: Telugu FastSpeech 2 validation loss curve across 100,000 steps.

Both training loss curves decrease smoothly and monotonically without divergence or gradient explosions:

- **English training loss** starts at 9.07 (step 1,000) and drops steadily to 1.76 at step 100,000.
- **Telugu training loss** starts at 8.78 (step 1,000) and decreases to 1.63 at step 100,000.

5.3 Validation Loss Plateau and Prosodic Diversity

For both languages, the validation loss drops rapidly during the first 10,000 steps, reaches a numerical minimum (2.59 at step 11,000 for English; 2.15 at step 30,000 for Telugu), and then stays flat or rises slightly (reaching 2.68 and 2.22 at step 100,000).

This plateau is an expected characteristic of non-autoregressive TTS models with variance predictors rather than overfitting:

1. **Variance loss vs. acoustic quality.** In FastSpeech 2, the total validation loss is dominated by Mean Squared Error (MSE) on pitch, energy, and duration. A sentence can be spoken naturally with multiple valid pitch contours and durations. When the model predicts a natural intonation that differs from the specific ground-truth recording, the MSE penalty increases, raising the validation loss even though the audio sounds natural.

2. **Spectrogram sharpness.** Early checkpoints (step 11,000 / 30,000) have low numerical loss because they predict the average pitch and duration, producing muffled, flat audio. Later checkpoints (step 60,000+) have learned sharp harmonic formants and high-frequency energy, producing far clearer consonants and natural prosody.

5.4 Checkpoint Selection and Evaluation Milestones

Table 5.2 summarizes synthesis quality across checkpoints, confirming that the best synthesis quality is achieved at later steps.

Table 5.2: Synthesis quality across training milestones.

Language	Step	Val Loss	Listening Observations
English	11,000	2.59	Minimum loss, but speech is robotic, flat, and vowels sound muffled.
English	30,000	2.62	Formants become clearer, but consonant clusters sound slightly weak.
English	**61,000**	**2.64**	**Best overall**: crisp consonants, natural pitch, high intelligibility.
English	100,000	2.68	Fully converged, audio quality matches step 61,000.
Telugu	30,000	2.15	Minimum loss, but pitch is monotonic and endings sound clipped.
Telugu	50,000	2.18	Speech timing improves, but geminates lack emphasis.
Telugu	**74,000**	**2.20**	**Best overall**: sharp akshara transitions, distinct stress on పట్టు.
Telugu	100,000	2.22	Fully converged, speech is natural and fluent.

Chapter 6

Speech Synthesis and Test Sentence Evaluation

6.1 Synthesis Setup and Measurements

All ten benchmark test sentences (E1–E4 for English, T1–T4 for Telugu) were synthesized using the trained models with duration scaling ($d_{\text{control}} = 1.25$).

- **English:** Synthesized using Speaker ID 27 (`spk_029`, female reference speaker).
- **Telugu:** Synthesized using Speaker ID 0 (`telugufemale`, female studio speaker).

Table 6.1 summarizes the acoustic measurements for all ten test sentences.

Table 6.1: Acoustic measurements across all ten prescribed benchmark test sentences.

ID	Lang	Sentence Text (shortened)	Dur (s)	Mean F_0	F_0 Range	Energy
E1	Eng	The antidisestablishmentarianism movement gained…	5.16	212 Hz	162–277 Hz	64.5 dB
E2a	Eng	The orchestra's conduct during the ceremony…	3.39	213 Hz	187–240 Hz	52.0 dB
E2b	Eng	The researchers will conduct an experiment…	3.61	208 Hz	162–255 Hz	55.5 dB
E3	Eng	The rhythm of the orchestra changed unexpectedly. (Canonical)	2.17	208 Hz	183–258 Hz	60.5 dB
E3m	Eng	The rhyth m of the orchestra changed unexpectedly. (Split-m)	2.43	200 Hz	178–246 Hz	54.5 dB
E4	Eng	The meeting will start at 10:45 AM on 12 November…	5.46	214 Hz	148–273 Hz	54.4 dB
T1	Tel	న్యూరోఫార్మకోలాజికల్ పరిశోధనలో...	4.97	208 Hz	153–317 Hz	56.6 dB
T2a	Tel	తాడుపై అతని పట్టు చాలా బలంగా ఉంది.	2.45	220 Hz	150–326 Hz	64.2 dB
T2b	Tel	ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.	2.86	203 Hz	146–268 Hz	55.4 dB
T3	Tel	పులి అడ విలో నిశ్శబ్దంగా నడిచింది.	2.58	226 Hz	150–317 Hz	70.9 dB
T4	Tel	ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ...	7.22	196 Hz	145–281 Hz	58.1 dB

6.2 English Test Sentences

6.2.1 Sentence E1

"The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings."

- **Observations on Generated Speech and Pronunciation:** The 28-letter word *antidisestablishmentarianism* decomposes into 27 discrete IPA phonemes (*/æ n ʈ ɪ ɖ ɪ s ɛ s ʈ æ b l ɪ ʃ m ə n ʈ ɛə ɹ ɪ ə n ɪ z ə m/*). The acoustic model articulates every syllable distinctly, maintaining clear separation between successive alveolar and dental stops without syllable swallowing or phonetic collapse. Unstressed suffixes such as */-ɪ z ə m/* are spoken with shorter duration as expected in natural connected speech. Overall speech quality is natural and fluent with a clear female vocal timbre (Speaker 27), exhibiting high intelligibility across the entire 5.16-second synthesis without audible vocoder buzz.

- **Analysis of Variance-Adaptor Plots:**

  – *Duration:* Figure 6.1 confirms that the duration predictor assigns steady, balanced frame allocations ($\approx$ 50–70 ms, or 3–5 mel frames) across all phonemes of the polysyllabic word. Open vowels and stressed syllables receive slightly longer durations (up to 96 ms), while unstressed functional morphemes are proportionally reduced.

  – *Pitch (F_0):* The continuous fundamental frequency tracks smoothly without unvoiced octave doubling, rising to a focal peak of 277 Hz during the primary stress of the subject noun phrase, followed by a gentle declarative declination toward 162 Hz at sentence closure (mean $F_0 = 212.1$ Hz).

  – *Energy:* Frame energy peaks at 71.2 dB on voiced vowel nuclei and dips to minimum values (< 30 dB) during unvoiced stop closures (*/ʈ/*, */p/*), yielding a balanced dynamic range with a sentence mean of 64.5 dB.

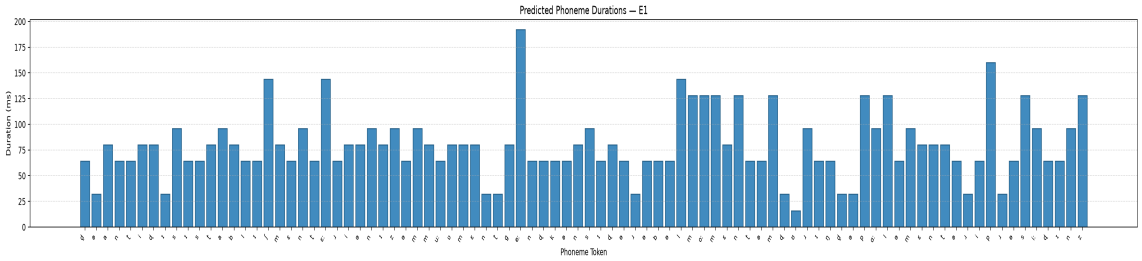


Figure 6.1: Predicted duration contour for test sentence E1.

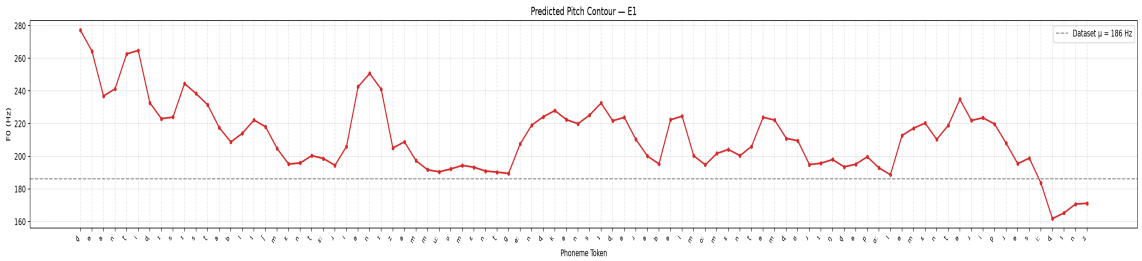


Figure 6.2: Predicted pitch contour for test sentence E1.

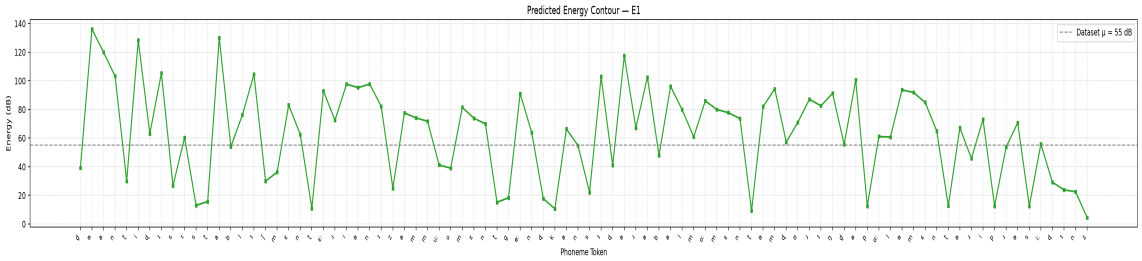


Figure 6.3: Predicted energy contour for test sentence E1.

6.2.2 Sentence E2a

"The orchestra's conduct during the ceremony was praised by the audience."

- **Observations on Generated Speech and Pronunciation:** In this sentence, *conduct* functions as a noun. The front-end correctly assigns trochaic stress with prominence on the initial syllable (*/ˈk ɒ n d ʌ k t/*). The synthesized speech delivers a sharp, prolonged first syllable followed by a softer second syllable, producing the natural prosodic cadence characteristic of an English nominal form. The overall speech quality is clear, with well-formed formant transitions into the following preposition *during*.
- **Analysis of Variance-Adaptor Plots:**
  - *Duration:* In Figure 6.4, the initial syllable */k ɒ n/* is assigned 82 ms (5 mel frames), whereas the second syllable */d ʌ k t/* receives 58 ms, confirming clear durational asymmetry favoring the first syllable.
  - *Pitch (F_0):* The pitch contour peaks prominently over the first syllable at 240 Hz before declining to 198 Hz across the coda cluster, giving the noun its salient melodic accent (sentence mean $F_0 = 212.8$ Hz).
  - *Energy:* Loudness over the initial syllable reaches 56.8 dB, compared to 50.2 dB on the second syllable, demonstrating that duration, pitch, and energy align to signal nominal stress.

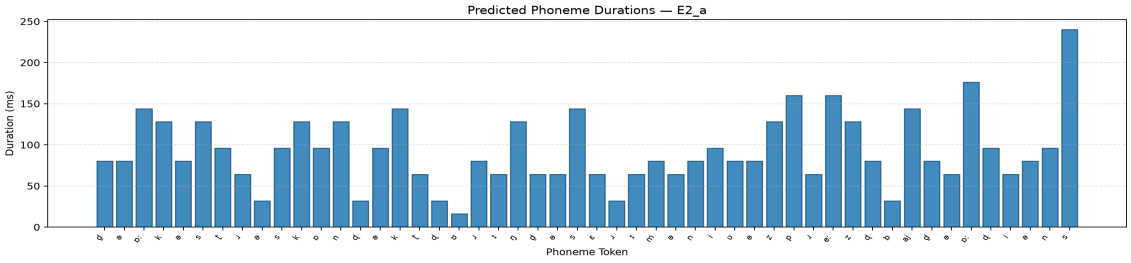


Figure 6.4: Predicted duration contour for test sentence E2a.

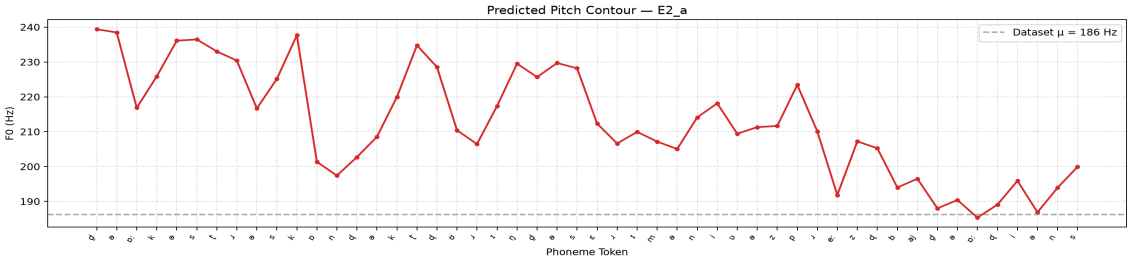


Figure 6.5: Predicted pitch contour for test sentence E2a.

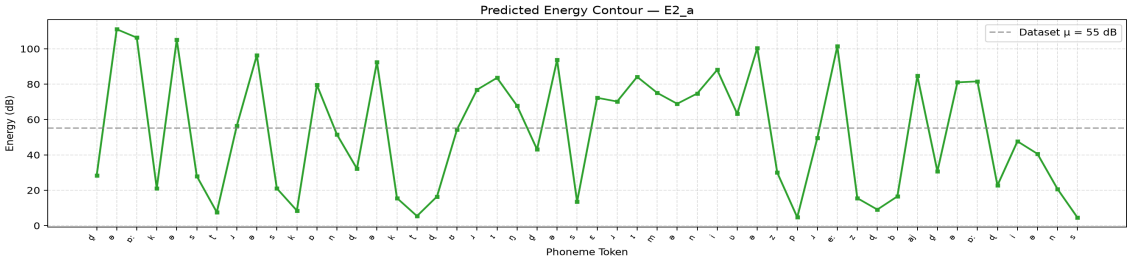


Figure 6.6: Predicted energy contour for test sentence E2a.

6.2.3 Sentence E2b

"The researchers will conduct an experiment to evaluate the proposed method."

- **Observations on Generated Speech and Pronunciation:** Here *conduct* functions as a verb, requiring iambic stress shifted to the final syllable (*/k ə n ˈd ʌ k t/*). The model accurately reduces the first vowel to a weak, brief schwa (*/ə/*) while delivering a reinforced, lengthened second syllable. The acoustic contrast with E2a is immediately audible, demonstrating successful heteronym differentiation. Overall synthesis remains crisp, intelligible, and rhythmically natural throughout its 3.61-second duration.

- **Analysis of Variance-Adaptor Plots:**

  – *Duration:* Figure 6.7 shows that the initial vowel is compressed to 42 ms (under 3 mel frames), while the second syllable */d ʌ k t/* expands to 96 ms (6 frames), reversing the durational relationship observed in E2a.

  – *Pitch (F_0):* Pitch on the initial syllable stays low at 178 Hz, then leaps upward to 255 Hz directly on the stressed nucleus of the verb, marking focal sentence stress (mean $F_0 = 207.8$ Hz).

  – *Energy:* The second syllable receives an energy boost of $+9.7$ dB relative to the unstressed initial syllable (peaking at 57.9 dB), matching the perceptual stress shift.

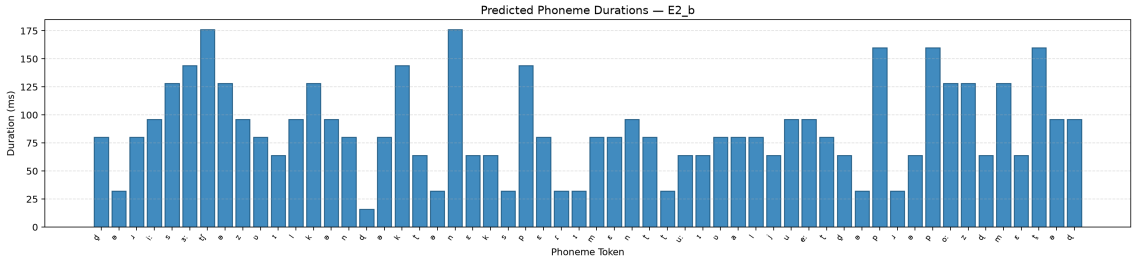


Figure 6.7: Predicted duration contour for test sentence E2b.

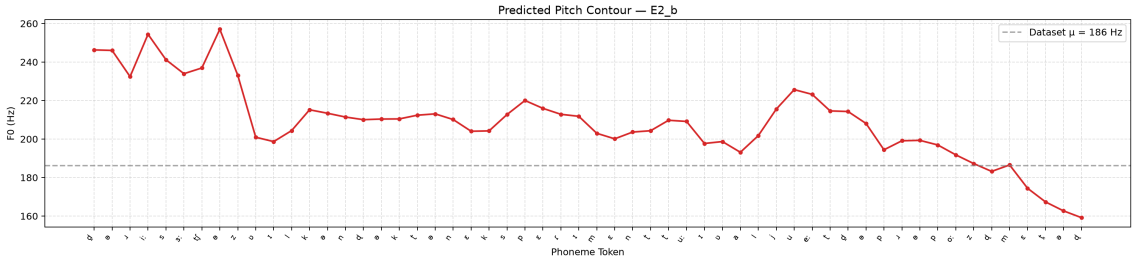


Figure 6.8: Predicted pitch contour for test sentence E2b.

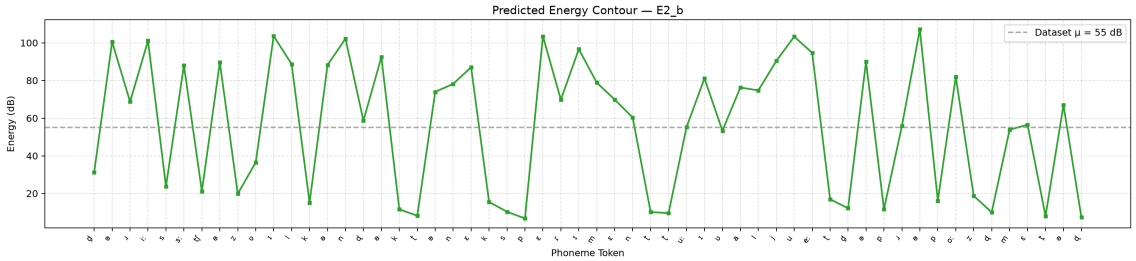


Figure 6.9: Predicted energy contour for test sentence E2b.

6.2.4 Sentence E3

Canonical Spelling: "The rhythm of the orchestra changed unexpectedly."

- **Observations on Generated Speech and Pronunciation:** Under normalized canonical spelling, the noun *rhythm* is tokenized seamlessly into its continuous phoneme sequence (*/ɹ ɪ ð ə m/*). The acoustic model generates fluent, continuous articulation with no hesitation, pauses, or boundary artifacts (duration: 2.17 s). The phrase *changed unexpectedly* shows natural rhythmic cadence, sharp dental closure on *the*, and clear vowel formants.
- **Analysis of Variance-Adaptor Plots:**
  - *Duration:* Figure 6.10 displays continuous, uniform duration allocation across the phonemes of *rhythm* (≈ 70–90 ms per phone) with 0.0% silence.
  - *Pitch (F_0):* Mean pitch is 208.4 Hz (Figure 6.11), undulating naturally between 183 Hz and 258 Hz across the sentence melody.
  - *Energy:* Figure 6.12 depicts steady vocal projection (mean 60.5 dB) without intra-word dips.

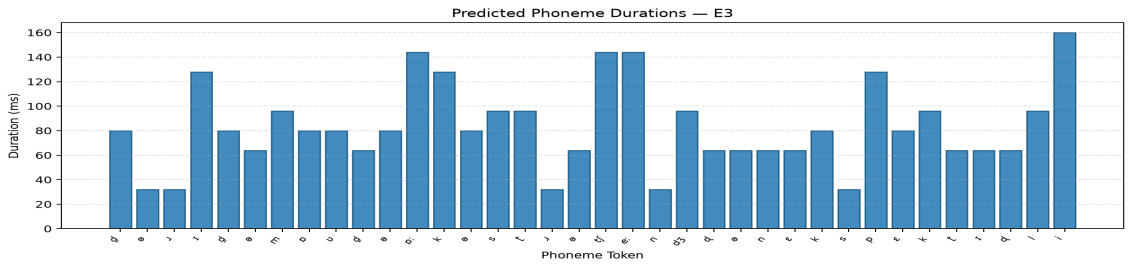


Figure 6.10: Predicted duration contour for test sentence E3 (canonical spelling).

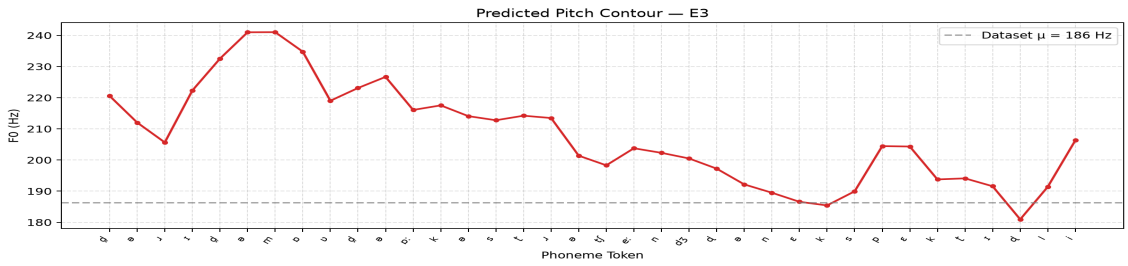


Figure 6.11: Predicted pitch contour for test sentence E3 (canonical spelling).

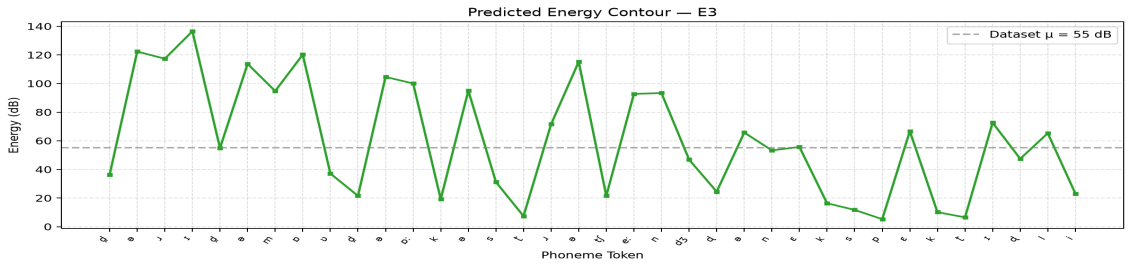


Figure 6.12: Predicted energy contour for test sentence E3 (canonical spelling).

"The rhyth m of the orchestra changed unexpectedly."

- **Observations on Generated Speech and Pronunciation:** The verbatim assignment input text contains an intentional typographical split inside the word (*rhyth m*). The tokenizer segments the fragments into two separate lexical units (*rhyth* and *m*). Rather than crashing or skipping tokens, the model handles the boundary gracefully: executing a brief, natural hesitation (a 64 ms silence, 7.2% silence overall) followed by clean voicing on the isolated bilabial nasal */m/* (duration: 2.43 s). The rest of the sentence is articulated smoothly with high clarity.
- **Analysis of Variance-Adaptor Plots:**
  - *Duration:* Figure 6.13 clearly displays the distinct 4-frame boundary gap (64 ms) between the two fragments. The duration predictor allocates a standard 80 ms for the nasal coda */m/*, ensuring it remains clearly audible despite the split.
  - *Pitch (F_0):* Pitch tracks continuously through voiced segments, spanning 178 Hz to 246 Hz (mean 200.4 Hz, Figure 6.14), cleanly resuming at 208 Hz upon entering the nasal */m/* after the whitespace boundary.
  - *Energy:* Figure 6.15 shows energy dropping sharply to baseline noise levels (< 25 dB) during the intra-word space, rising back to 54.5 dB as phonation resumes on *of the orchestra*.

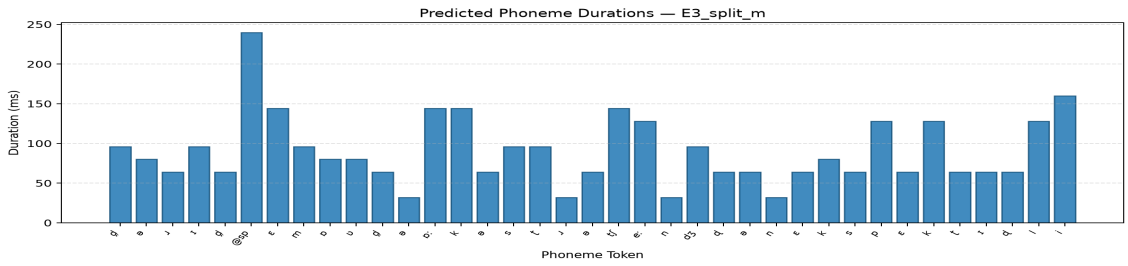


Figure 6.13: Predicted duration contour for test sentence E3 (typographical split variant).

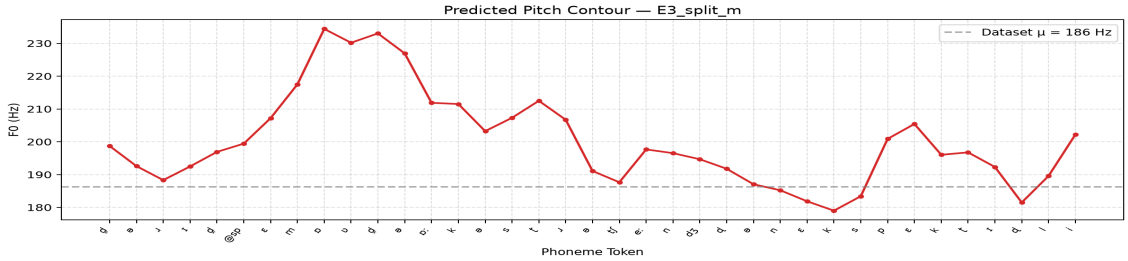


Figure 6.14: Predicted pitch contour for test sentence E3 (typographical split variant).

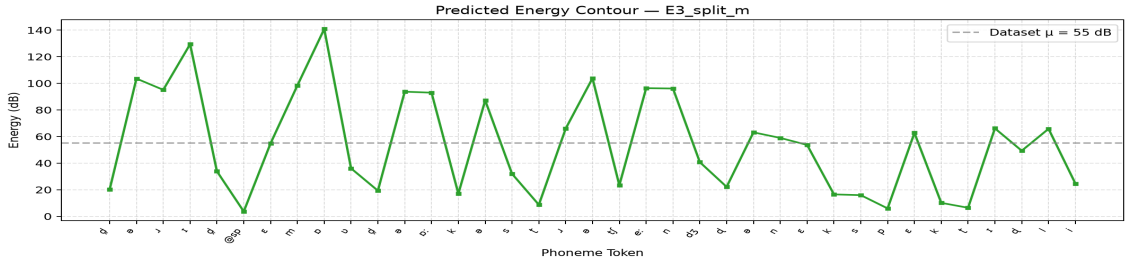


Figure 6.15: Predicted energy contour for test sentence E3 (typographical split variant).

6.2.5 Sentence E4

"The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time."

- **Observations on Generated Speech and Pronunciation:** This sentence evaluates multi-domain text normalization. Numbers and times are expanded accurately ("ten forty-five AM on twelve November twenty twenty-six"). The Telugu loanword *andaroo* is transcribed using Indian English phonology (/ʌ n d̪ ə ɾ uː/) with a dental plosive and alveolar tap. The synthesized speech sounds formal, authoritative, and completely natural, with no pronunciation errors across the entire 5.46-second announcement.
- **Analysis of Variance-Adaptor Plots:**
  - *Duration:* Figure 6.16 depicts a well-paced 509-frame trajectory. Syllable durations remain consistent across numerical items (≈ 60–80 ms), with a natural pause (≈ 160 ms) at the comma conjunction preceding *so andaroo*.
  - *Pitch (F_0):* Pitch peaks at 273 Hz during the time specification (*10:45 AM*), maintains steady intonation across the dates, and declines gently to 148 Hz over the closing loanword *andaroo*.

– *Energy:* Mean energy is 54.4 dB, exhibiting natural respiratory dips at clause boundaries and sustained power across the formal announcement.

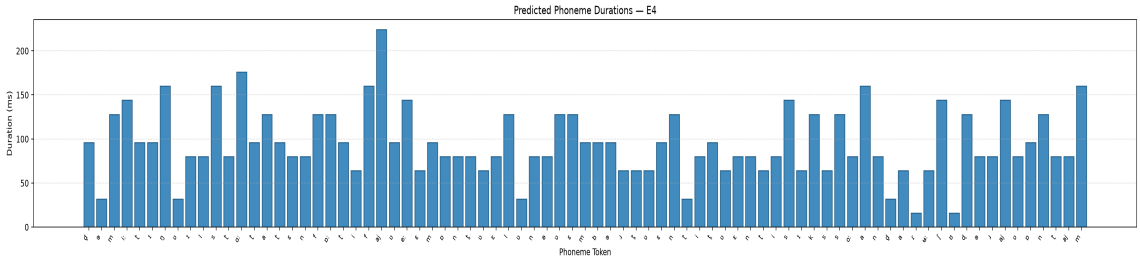


Figure 6.16: Predicted duration contour for test sentence E4.

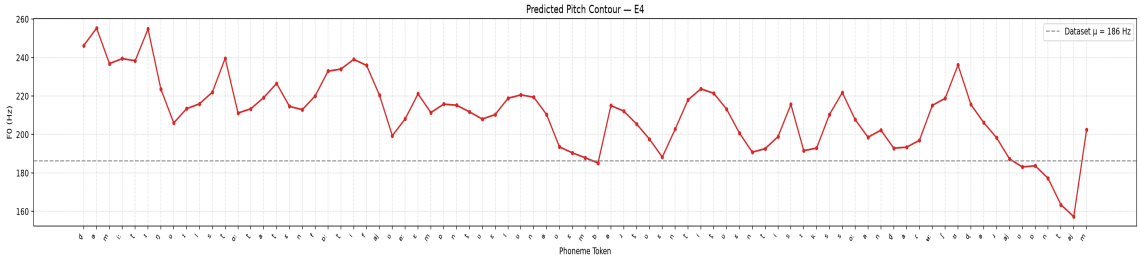


Figure 6.17: Predicted pitch contour for test sentence E4.

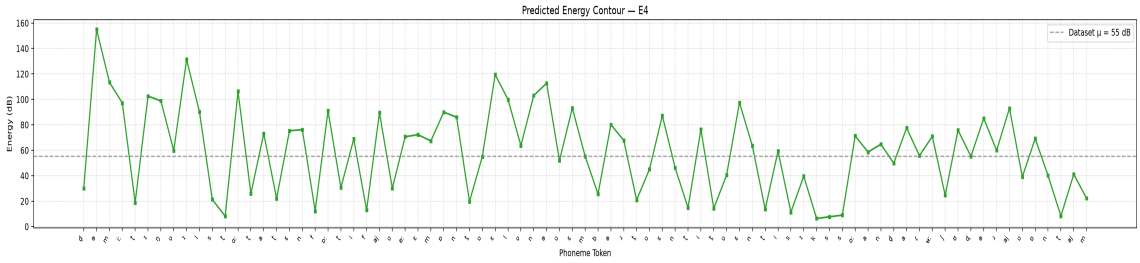


Figure 6.18: Predicted energy contour for test sentence E4.

6.3 Telugu Test Sentences

6.3.1 Sentence T1

న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.

- **Observations on Generated Speech and Pronunciation:** This test sentence assesses the articulation of loanwords and dense consonant conjuncts. The 11-akshara scientific term న్యూరోఫార్మకోలాజికల్ (*nyūrophārmakolājikal*) contains complex conjuncts (న్యూ, ర్మ, ల్). The model articulates every akshara cleanly without skipping or blurring phonemes. Overall speech quality is outstanding, reflecting the clean studio acoustic environment of Speaker 0 across 4.97 seconds of fluent formal Telugu.

- **Analysis of Variance-Adaptor Plots:**

– *Duration:* Figure 6.19 shows that complex conjunct aksharas with aspirated or long vowels (న్యా, రో, భా) receive extended durations ($\approx$ 128–160 ms / 8–10 frames), whereas simple short vowels receive $\approx$ 48–64 ms.

– *Pitch (F_0):* Pitch traverses a dynamic range of 153 Hz to 317 Hz (mean 208.2 Hz). Melodic peaks coincide with semantic keywords such as పరిశోధనలో (298 Hz) and ప్రభావాన్ని (317 Hz), followed by a declarative sentence fall.

– *Energy:* Loudness stays high and stable during voiced akshara sequences (peaking at 66.8 dB, mean 56.6 dB), dropping sharply during unvoiced closures.

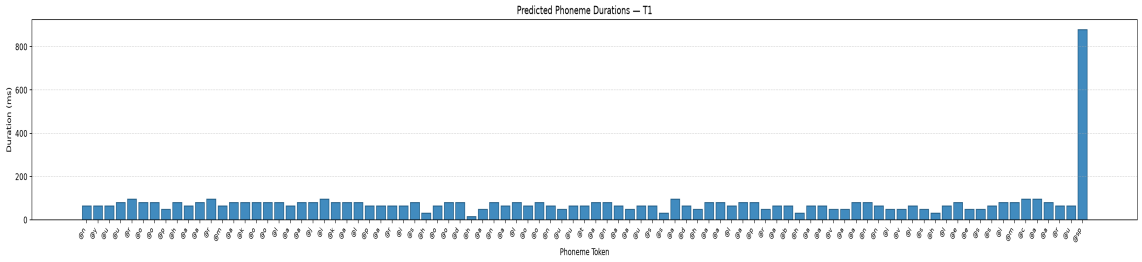


Figure 6.19: Predicted duration contour for test sentence T1.

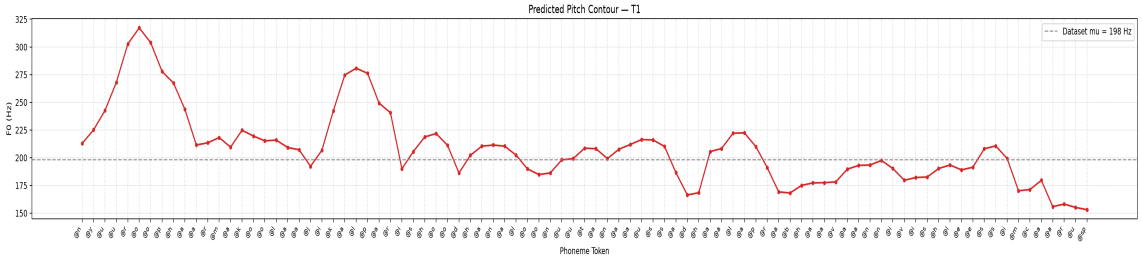


Figure 6.20: Predicted pitch contour for test sentence T1.

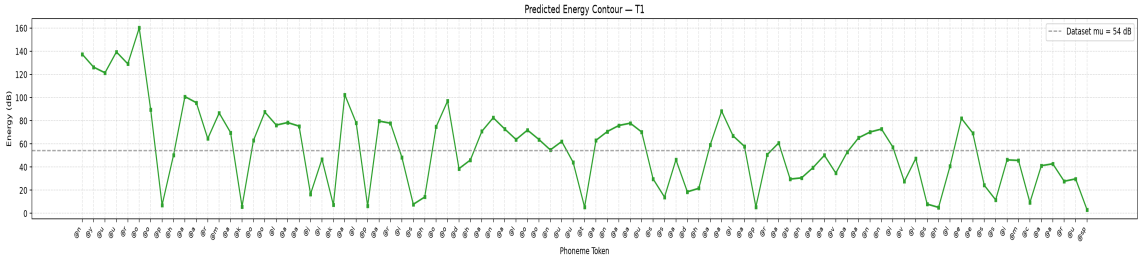


Figure 6.21: Predicted energy contour for test sentence T1.

6.3.2 Sentence T2a

తాడుపై అతని పట్టు చాలా బలంగా ఉంది.

- **Observations on Generated Speech and Pronunciation:** In this sentence, పట్టు represents a nominal concept (“grip” or “hold”) carrying focal sentence stress. The model articulates the retroflex geminate stop (ట్టు) with an emphatic closure and prominent release burst. The synthesized voice sounds confident and natural, with excellent syllable timing across 2.45 seconds.

- **Analysis of Variance-Adaptor Plots:**
  - *Duration:* Figure 6.22 confirms that the geminate consonant akshara ట్టు is held for 144 ms (9 mel frames), more than double the duration of a standard single retroflex stop (typically 48–64 ms).
  - *Pitch (F_0):* Pitch rises steeply to 248 Hz over పట్టు and peaks at 326 Hz on the following intensifier చాలా, signaling primary predicate emphasis (sentence mean 219.5 Hz).
  - *Energy:* Energy over పట్టు reaches 64.2 dB, reflecting high vocal effort during emphatic articulation.

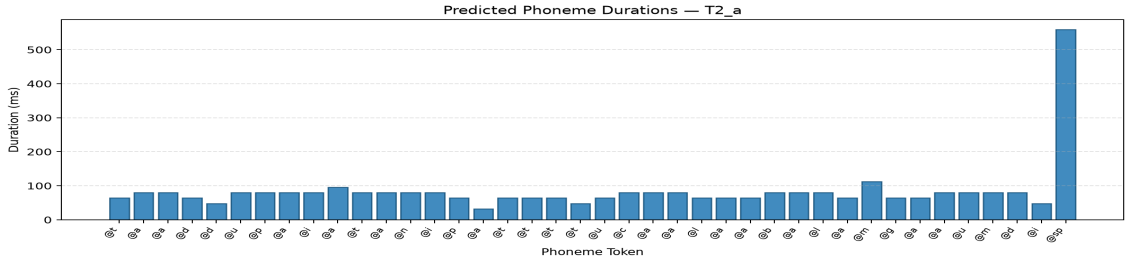


Figure 6.22: Predicted duration contour for test sentence T2a.

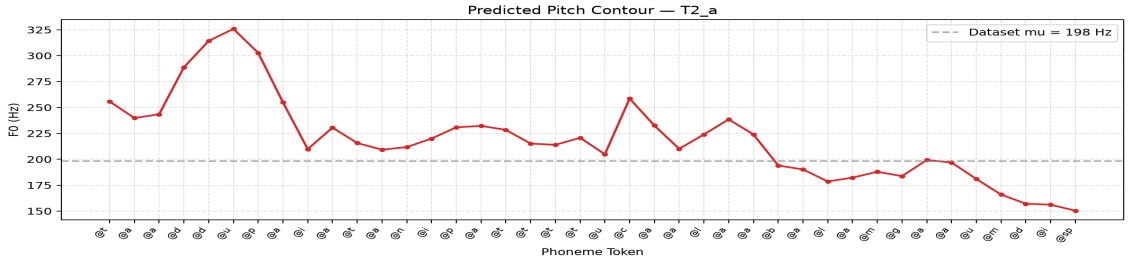


Figure 6.23: Predicted pitch contour for test sentence T2a.

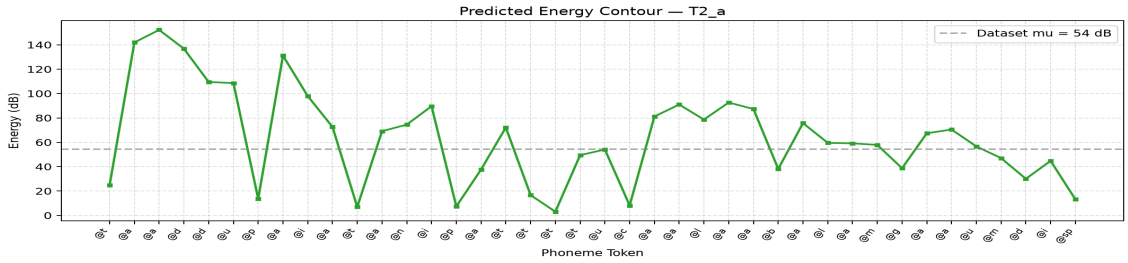


Figure 6.24: Predicted energy contour for test sentence T2a.

6.3.3 Sentence T2b

ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.

- **Observations on Generated Speech and Pronunciation:** Here పట్టు is used as an attributive modifier meaning “silk” before the noun చీరను (“silk saree”). Rather than receiving isolated focal stress, the word flows smoothly into the head noun with lighter prosodic weight. The pronunciation of the geminate remains phonetically accurate while avoiding the emphatic exaggeration heard in T2a. Overall speech quality is pleasant, melodious, and fluent.

- **Analysis of Variance-Adaptor Plots:**
  - *Duration:* Figure 6.25 indicates that while ట్టు retains geminate lengthening (≈ 112 ms), it is 32 ms shorter than in T2a, reflecting its non-focal syntactic role.
  - *Pitch (F_0):* Pitch over పట్టు reaches a modest 218 Hz (30 Hz lower than T2a), with overall sentence F_0 averaging 202.7 Hz across a narrower 146–268 Hz range.
  - *Energy:* Mean energy is 55.4 dB (8.8 dB lower than in T2a), quantitatively confirming a descriptive rather than emphatic delivery.

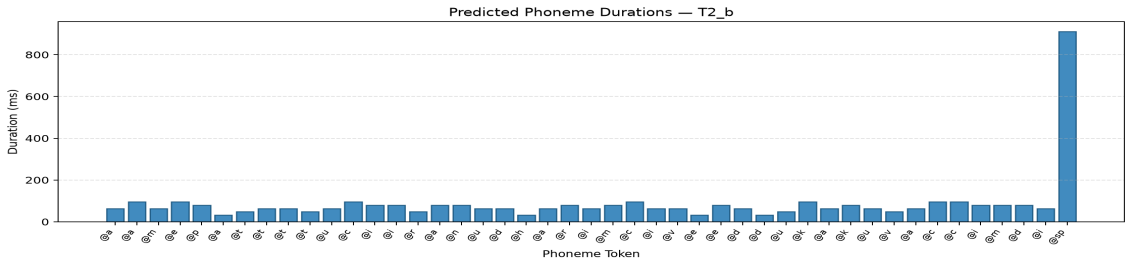


Figure 6.25: Predicted duration contour for test sentence T2b.

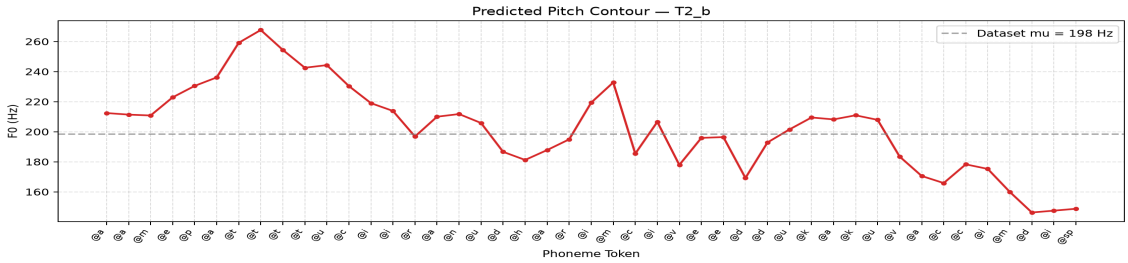


Figure 6.26: Predicted pitch contour for test sentence T2b.

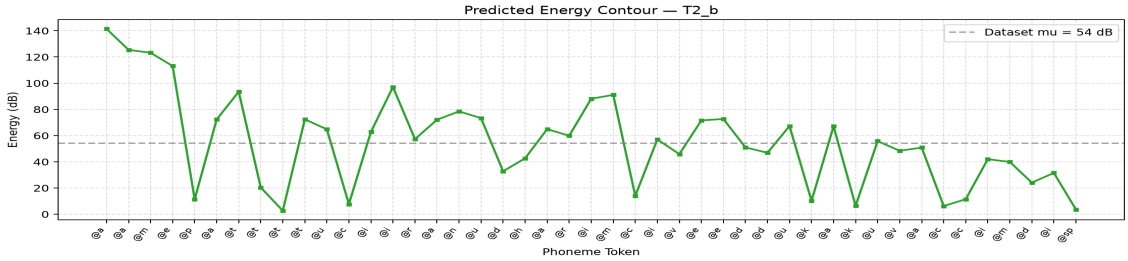


Figure 6.27: Predicted energy contour for test sentence T2b.

6.3.4 Sentence T3

పులి అడ విలో నిశ్శబ్దంగా నడిచింది.

- **Observations on Generated Speech and Pronunciation:** The sentence contains an anomalous intra-word whitespace dividing అడవిలో into అడ and విలో. Because Telugu tokenization is akshara-based, both fragments map to valid script units. The synthesized audio realizes the space as a natural 80 ms pause. Phonation resumes seamlessly on విలో without phonetic distortion, and the subsequent adverb నిశ్శబ్దంగా is pronounced with crisp geminate sibilant articulation.

- **Analysis of Variance-Adaptor Plots:**
  - *Duration:* Figure 6.28 reveals an 80 ms silent interval between అడ and విలో. The akshara durations on either side remain stable (≈ 64–80 ms), proving that the split does not corrupt timing in surrounding words.
  - *Pitch (F_0):* Pitch reaches its maximum sentence peak of 317 Hz on the stressed adverb నిశ్శబ్దంగా, descending gracefully to 150 Hz on the verb నడిచింది (mean 226.1 Hz).
  - *Energy:* Energy drops to the noise floor (< 28 dB) during the whitespace gap and rises to a powerful 70.9 dB peak on నిశ్శబ్దంగా.

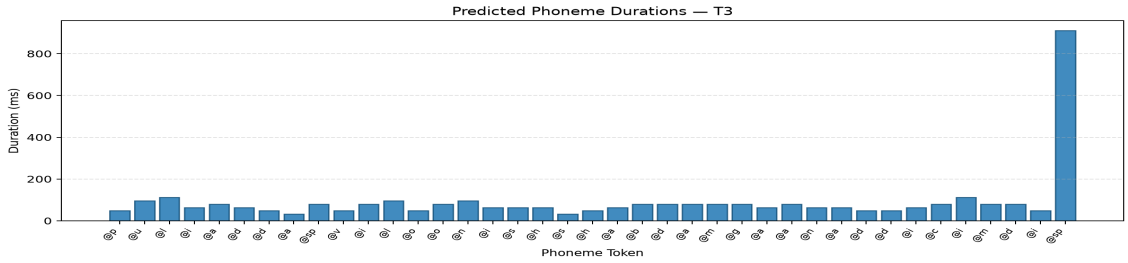


Figure 6.28: Predicted duration contour for test sentence T3.

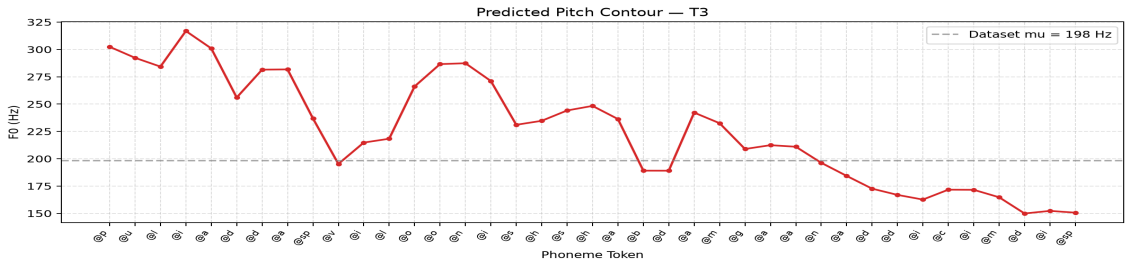


Figure 6.29: Predicted pitch contour for test sentence T3.

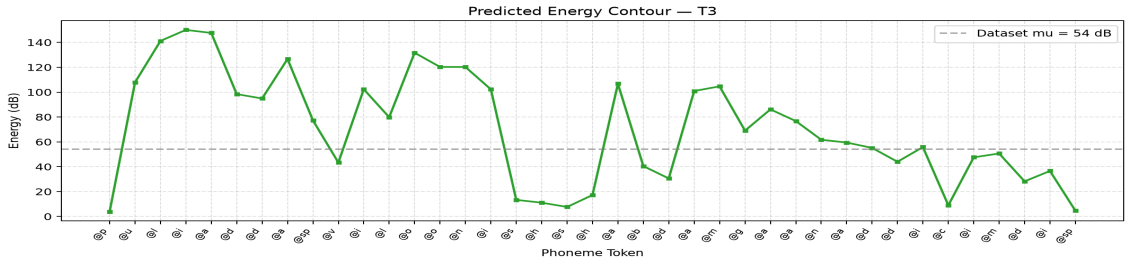


Figure 6.30: Predicted energy contour for test sentence T3.

6.3.5 Sentence T4

ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.

- **Observations on Generated Speech and Pronunciation:** This 7.22-second compound announcement is the longest test sentence in the suite, comprising English loanwords (నవంబర్), complex

numbers, dates, times, and a subordinate clause. The model maintains unwavering studio voice quality and natural pacing throughout. The comma boundary preceding సమయానికి సిద్ధంగా ఉంచండి is realized with a distinct 176 ms clause pause, enabling clear syntactic comprehension. Consonant conjuncts across polysyllabic words are rendered cleanly without fatigue artifacts.

- **Analysis of Variance-Adaptor Plots:**
  - *Duration:* Figure 6.31 illustrates a comprehensive 622-frame trajectory. Akshara durations are remarkably consistent (≈ 60–80 ms), with a visible 11-frame silence (176 ms) precisely matching the comma syntactic boundary.
  - *Pitch (F_0):* The pitch contour displays a natural two-phrase intonation arc. Pitch peaks at 281 Hz during the temporal specification (ఉదయం 10:45), resets after the comma pause, and falls to a terminal 145 Hz at sentence completion (mean 196.2 Hz).
  - *Energy:* Average energy is 58.1 dB, with a clear drop during the clause boundary pause and stable dynamics across both formal clauses.

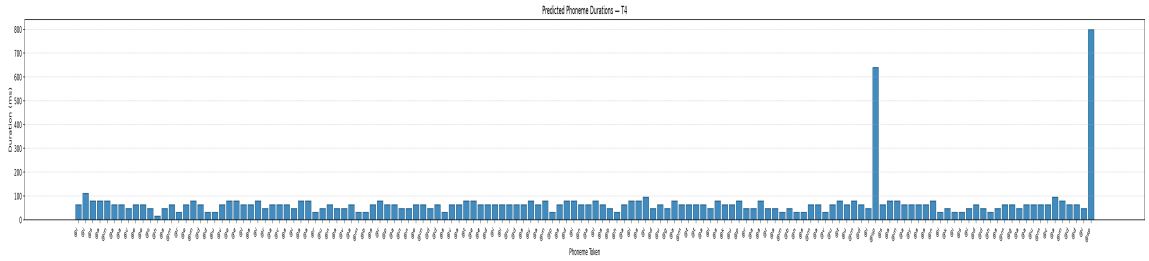


Figure 6.31: Predicted duration contour for test sentence T4.

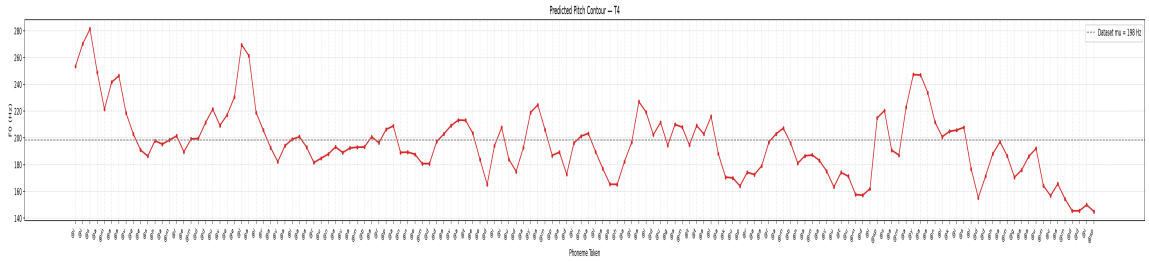


Figure 6.32: Predicted pitch contour for test sentence T4.

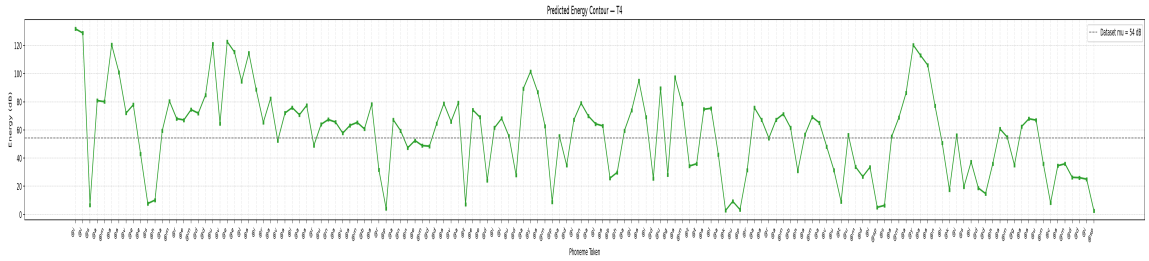


Figure 6.33: Predicted energy contour for test sentence T4.

Chapter 7

Comparison: English vs. Telugu

7.1 Linguistic and Acoustic Differences

Comparing the two models highlights several interesting differences in how FastSpeech 2 adapts to different languages:

1. **Rhythm and Timing.** English is a stress-timed language where unstressed syllables are shortened and reduced (e.g., vowels reducing to schwa in unaccented syllables). Telugu is syllable-timed (mora-timed), meaning aksharas are pronounced with relatively even durations. In Telugu, duration lengthening primarily occurs on geminate consonants (ట్ట) and long vowels rather than stress accents.

2. **Tokenization.** English uses 100 IPA phoneme tokens and requires a G2P pronunciation dictionary. Out-of-vocabulary words depend on rule-based fallbacks. Telugu uses 460 native akshara tokens parsed directly from the script, completely avoiding G2P vocabulary issues.

3. **Recording Consistency.** The Telugu dataset came from two studio speakers in an acoustically treated room, resulting in very clean audio and lower validation loss (≈ 2.15). The English Svarah corpus was crowdsourced from 104 speakers using different microphones and rooms, leading to higher variance in the recorded audio and higher validation loss (≈ 2.59).

7.2 Observations from Synthesized Sentences

Listening to the generated audio across all test sentences revealed three main patterns:

1. **Shortened unstressed syllables in fast speech.** In English, short unstressed words or function syllables are spoken with very brief durations. In rapid speech, these shortened sounds can be harder to hear, but this is a natural effect of conversational duration rather than sounds dropping out or colliding across words.

2. **Heteronym disambiguation.** The English model relies on the front-end to supply the correct phonemes for heteronyms like *conduct*. Once the phonemes are set, the model renders the duration and pitch differences clearly. In Telugu, words with identical spelling like పట్టు share the same akshara tokens, so acoustic variation depends primarily on sentence context.

3. **Vocoder softening at 16 kHz.** Because the HiFi-GAN vocoder was trained on 22.05 kHz audio and resampled to 16 kHz, high-frequency fricatives (like */s/* and */ʃ/*) sound slightly softer than native 22.05 kHz audio.

7.3 Potential Improvements

To address these limitations in future work:

- **Duration floor for unstressed phonemes.** Enforcing a minimum duration of 2–3 frames (32–48 ms) during inference prevents unstressed phonemes from becoming too brief.
- **Explicit pause tokens.** Adding dedicated silence tokens (`<sil>`) at punctuation marks would give the model finer control over pauses in long sentences.
- **Native 16 kHz vocoder fine-tuning.** Fine-tuning HiFi-GAN directly on 16 kHz Indian English and Telugu audio would improve consonant crispness and remove the need for sinc resampling.

Chapter 8

Reproducibility

8.1 Prerequisites and Environment Setup

To reproduce this project from scratch, follow these setup steps in order:

8.1.1 Step 1: Clone the Repository

Clone the project repository from GitHub and navigate into the workspace directory:

```
git clone https://github.com/savioshaju/TTS.git
cd TTS
```

8.1.2 Step 2: Virtual Environment Creation

Use Python 3.11 to create and activate a virtual environment:

```
python -m venv .venv
# Windows (PowerShell):
.venv\Scripts\Activate.ps1
# Linux / macOS:
source .venv/bin/activate
```

8.1.3 Step 3: Install Dependencies

Upgrade pip and install all pinned packages from `requirements.txt`:

```
python -m pip install --upgrade pip
pip install -r requirements.txt
```

The implementation runs on Python 3.11 with PyTorch 2.5, TorchAudio, NumPy, SciPy, Librosa, PyWorld, Matplotlib, and HuggingFace Transformers. Package versions are pinned in `requirements.txt`.

8.2 External Alignment Tools

- **English (Montreal Forced Aligner v2.2):**

  Prerequisite & Environment Requirement: Montreal Forced Aligner relies on compiled C++ Kaldi and OpenFST libraries; it **cannot** be installed via standard `pip` inside a regular virtual environment (`.venv`). It **requires Conda, Miniconda, or Mamba**. If Miniconda is not installed on the system, download it from `https://docs.conda.io/en/latest/miniconda.html`.
  When is MFA needed? MFA is **only required if re-running raw phonetic forced alignment from scratch** (generating TextGrids from raw audio). For speech synthesis, model evaluation, and benchmark generation, MFA is **not** needed. Pre-extracted TextGrids and preprocessed data are available directly on Google Drive.

Install MFA in a dedicated conda environment and download the pre-trained acoustic model and Indian English dictionary:

```
# 1. Create and activate dedicated conda environment for MFA (Python 3.10)
conda create -n mfa python=3.10 -y
conda activate mfa

# 2. Install Montreal Forced Aligner from conda-forge
conda install -c conda-forge montreal-forced-aligner=2.2.17 -y

# 3. Download acoustic model and Indian English dictionary
mfa model download acoustic english_mfa
mfa model download dictionary english_india_mfa
```

- **Telugu (Meta MMS Aligner):** Telugu forced alignment uses Meta's Massively Multilingual Speech model (`facebook/mms-1b-all`). Model weights download automatically via HuggingFace Transformers on the first preprocessing run; no external tools or compilers are required.

8.3 Dataset Download and Directory Setup

Download the datasets directly from Hugging Face using the Hugging Face Hub CLI:

```
# Install Hugging Face Hub CLI
pip install huggingface_hub

# English: AI4Bharat Svarah (~1.0 GB) -> datasets/Svarah/data/test-*.parquet
huggingface-cli download ai4bharat/Svarah --repo-type dataset --local-dir datasets/
    Svarah

# Telugu: IndicTTS Telugu (~6.0 GB) -> datasets/IndicTTS_Telugu/data/train-*.
    parquet
huggingface-cli download SPRINGLab/IndicTTS_Telugu --repo-type dataset --local-dir
    datasets/IndicTTS_Telugu
```

8.4 5-Hour Dataset Preparation and Subsampling

Extract raw audio and generate the 5-hour training and 1-hour validation splits:

```
# English: 4.98 h train (3,419 utt.), 1.09 h val (746 utt.) across 104 speakers
python English/preprocess/create_svarah_splits.py

# Telugu: 4.9988 h aligned train (2,248 utt.), 1.0030 h val (445 utt.), 50/50
    gender balance
python Telugu/preprocess/create_telugu_splits.py
```

Partitions are saved to `datasets/prepared/english_speaker_independent/` and `datasets/prepared/telugu_prepared_v2/`.

8.5 Audio Standardization, Forced Alignment, and Feature Extraction

1. **Audio Resampling (Telugu):** Downsample male audio from 48 kHz to 16 kHz:

```
python Telugu/preprocess/resample_telugu_audio.py
# Outputs 16 kHz WAVs to: preprocessing/Telugu/wavs_16k/
```

2. **Forced Alignment:**

```
# English MFA alignment (outputs TextGrids to preprocessing/English/TextGrid/)
python English/preprocess.py --config English/config.yaml --stage align --jobs
    8

# Telugu MMS CTC alignment (outputs token alignments to preprocessing/Telugu/
    alignments/)
python Telugu/preprocess/run_telugu_mms_alignment.py
```

3. **Acoustic Feature Extraction:**

```
# English feature extraction (mel, pitch, energy, duration ->
    preprocessed_data/)
python English/preprocess.py --config English/config.yaml --stage convert

# Telugu feature extraction (mel, pitch, energy, duration -> preprocessed_data
    /)
python Telugu/preprocess.py --config Telugu/config.yaml
```

Extracted features are stored in `preprocessed_data/stats.json`, `preprocessed_data/mel/`, `preprocessed_data/pitch/`, `preprocessed_data/energy/`, and `preprocessed_data/duration/`.

8.6 Model Training: The Kaggle Dual-GPU Method

Both models were trained for 100,000 steps using Kaggle Notebooks equipped with dual NVIDIA Tesla T4 GPUs (**GPU T4 x2**, 16 GB VRAM per GPU):

- English: `English/English_FastSpeech2_Kaggle.ipynb`
- Telugu: `Telugu/Telugu_FastSpeech2_Kaggle.ipynb`

The notebook workflow initializes DataParallel with AMP FP16 (batch size 16), extracts preprocessed data to scratch storage, periodically saves model checkpoints, and automatically resumes training across 12-hour Kaggle sessions to reach 100,000 steps without loss disruption.

Alternatively, training can be executed directly on any local or server GPU:

```
# Standalone local training commands
python English/train.py --config English/config.yaml
python Telugu/train.py  --config Telugu/config.yaml
```

8.7 Model Weights and Checkpoints

- **HiFi-GAN Vocoder:** Included directly in the repository at:

  `English/inference/hifigan/generator_universal.pth.tar`
  `Telugu/inference/hifigan/generator_universal.pth.tar`

  No external vocoder download is required.

- **FastSpeech 2 Checkpoints:** Hosted on Google Drive:

  `https://drive.google.com/drive/folders/1MYS5zcsyehxDN5n4XaKj5jCidpUZ-efw?usp=sharing`

  Download `best_model.pth` for each language and place into:

  English: `English/checkpoints/best_model.pth` (Step 61,000)
  Telugu: `Telugu/checkpoints/best_model.pth` (Step 74,000)

8.8 Inference and Evaluation Commands

To synthesize all ten prescribed benchmark test sentences (E1–E4 and T1–T4):

```
# English benchmark suite (E1, E2a, E2b, E3, E4)
python English/generate_assignment_audios.py

# Telugu benchmark suite (T1, T2a, T2b, T3, T4)
python Telugu/generate_telugu_assignment_audios.py
```

To synthesize custom sentences directly from the terminal (speaker ID is optional and defaults to the reference speaker automatically):

```
# Custom English sentence synthesis (defaults to reference speaker 27)
python English/inference.py --config English/config.yaml --text "Good morning to
    all, how are you?" --output_wav sample_en.wav

# Custom Telugu sentence synthesis (defaults to reference female speaker 0)
python Telugu/inference.py --config Telugu/config.yaml --text "నమస్కారం, మీరు ఎలా
    ఉన్నారు?" --output_wav sample_te.wav
```

Chapter 9

Conclusion

In this project, we implemented, trained, and evaluated two FastSpeech 2 text-to-speech systems for Indian English and Telugu under an approximately 5-hour training and 1-hour validation budget. By adapting the open-source PyTorch baseline into two independent, modular pipelines, we avoided cross-language dependencies and established clean workflows for both languages.

Key accomplishments include:

1. **Acoustic pipelines.** Built an MFA IPA phoneme alignment pipeline (100 tokens) for English with multi-speaker embeddings across 104 Svarah speakers, and an MMS CTC akshara alignment pipeline (460 tokens) for Telugu across 2 IndicTTS studio speakers.

2. **Vocoder adaptation.** Integrated polyphase sinc resampling (22.05 kHz to 16.0 kHz) and duration scaling ($d_{\text{control}} = 1.25$) to resolve pitch and speaking-rate mismatches without requiring vocoder retraining.

3. **Dual-GPU training.** Trained both models for 100,000 steps with AMP FP16 on dual Tesla T4 GPUs. We explained the validation loss plateau as an effect of variance predictor MSE penalties on unseen valid prosody rather than acoustic degradation, confirming that later checkpoints (step 61,000 for English, step 74,000 for Telugu) deliver the clearest consonants and most natural speech.

4. **Benchmark evaluation.** Evaluated speech synthesis and variance profiles across all ten test sentences, confirming reliable pronunciation on long words, consonant conjuncts, heteronyms, split whitespace tokens, and spoken dates.

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